INCH-POUND
MIL-DTL-17840D(SH)
29 January 2018
SUPERSEDING
MIL-P-17840C(SH)
10 March 1986

DETAIL SPECIFICATION

PUMPS, CENTRIFUGAL, CLOSE-COUPLED, NAVY STANDARD (FOR SURFACE SHIP APPLICATION)

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers alternating current (AC) electric motor driven, Navy standard, close-coupled, single-stage, centrifugal pumps for surface ship application.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-N-281	-	Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections
QQ-N-286	-	Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500)
TT-P-645	-	Paint, Alkyd Type, Formula Number 84

FEDERAL STANDARDS

FED-STD-H28	-	Screw-Thread Standards for Federal Services
FED-STD-H28/2	-	Screw-Thread Standards for Federal Services Section 2 Unified Inch Screw Threads – UN and UNR Thread Forms

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-DTL-1222	-	Studs, Bolts, Screws and Nuts for Applications Where a High Degree of Reliability is Required; General Specification for
MIL-DTL-15024	-	Plates, Tags, and Bands for Identification of Equipment, General Specification for
MIL-DTL-15024/5	-	Plates, Identification
MIL-DTL-15090	-	Enamel, Equipment, Light Gray (Navy Formula No. 111)
DOD-P-15328	-	Primer (Wash), Pretreatment (Formula No.117 for Metals) (Metric)
MIL-C-15726	-	Copper-Nickel Alloy, Sheet, Plate, Strip, Bar, Rod, and Wire
MIL-T-16420	-	Tube, Copper-Nickel Alloy, Seamless and Welded (Copper Alloy Numbers 715 and 706)
MIL-M-17059	-	Motors, 60-Cycle, Alternating-Current, Fractional H.P. (Shipboard Use)
MIL-DTL-17060	-	Motors, Alternating Current, Integral-Horsepower, Shipboard Use
MIL-P-17416	-	Packing Material, Metallic, Flexible (Symbol 1430)
DOD-G-24508	-	Grease, High Performance, Multipurpose (Metric)
MIL-DTL-24696	-	Gasket, Sheet, Non-Asbestos
MIL-C-24723	-	Castings, Nickel-Copper Alloy
MIL-PRF-24790	-	Packing Material, Braided, Non-Asbestos

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-777	-	Schedule of Piping, Valves, Fittings, and Associated Piping Components for Naval Surface Ships
DOD-STD-1399-301	-	Interface Standard for Shipboard Systems, Ship Motion and Attitude (Metric)
MIL-STD-1472	-	Human Engineering
MIL-STD-1474	-	Noise Limits

(Copies of these documents are available online at http://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

MILITARY SPECIFICATION SHEETS

MS18229	-	Plug for O-Ring Gasket

(Copies of this document are available online at http://quicksearch.dla.mil/.)

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NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

B-214	-	Root Connections for Attaching Piping
803-1385850	-	Piping, Instrument, Pressure for All Services

(Copies of these documents are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of these documents may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AR-GIB-010/278	-	Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery
S9086-CM-STM-020/078	-	NSTM Chapter 078, Volume 2 – Gaskets and Packing
T9070-AL-DPC-020/077-2	-	NAVSEA Hazardous Material Avoidance Process

(Copies of these documents are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN BEARING MANUFACTURES ASSOCIATION (ABMA)

ANSI/ABMA 9	-	Load Ratings and Fatigue Life for Ball Bearings

(Copies of this document are available online at www.americanbearings.org.)

ASTM INTERNATIONAL

ASTM A494/A494M	-	Standard Specification for Castings, Nickel and Nickel Alloy
ASTM B61	-	Standard Specification for Steam or Valve Bronze Castings
ASTM B164	-	Standard Specification for Nickel-Copper Alloy Rod, Bar, and Wire
ASTM B369	-	Standard Specification for Copper-Nickel Alloy Castings
ASTM B584	-	Standard Specification for Copper Alloy Sand Castings for General Applications
ASTM D5363	-	Standard Specification for Anaerobic Single-Component Adhesives (AN)
ASTM F1511	-	Standard Specification for Mechanical Seals for Shipboard Pump Applications

(Copies of these documents are available online at www.astm.org.)

HYDRAULIC INSTITUTE (HI)

HI 40.6	-	Methods for Rotodynamic Pump Efficiency Testing

(Copies of this document are available online at www.pumps.org.)

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NSF INTERNATIONAL

NSF/ANSI 61 - Drinking Water System Components - Health Effects

(Copies of this document are available online at www.nsf.org.)

SAE INTERNATIONAL

SAE J1926/1 - Connections for General Use and Fluid Power - Ports and Stud Ends with ASME B1.1 Threads and O-Ring Sealing - Part 1: Threaded Port with O-Ring Seal in Truncated Housing

SAE AS35411 - Fittings, Lubrication

(Copies of these documents are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 General design. The design and construction of pumps shall comply with the space and weight requirements as specified (see 6.2). The pump unit shall perform under the list, pitch, roll, and trim conditions in accordance with DOD-STD-1399-301. Pumps shall operate continuously at any point over the entire flow range and the design conditions as specified (see 6.2).

3.2.1 Reliability. The requirements of this specification are imposed exclusively to obtain equipment of utmost reliability for the service intended. The assurance of maximum reliability is the paramount controlling principle in the design, fabrication, assembly, and testing of this equipment.

3.2.2 Operating environment. The equipment specified herein will be operated, maintained, and repaired on board Navy ships. Attention is directed to the high equipment density of shipboard machinery spaces, due to the fact that maintenance and repair will be made underway in heavy seas, equipment will operate unattended, and maintenance personnel may not be seasoned mechanics. The requirements for maximum reliability directly relate to those shipboard environmental and service conditions, and they shall be fully considered in the pump design. The aspects of human engineering in accordance with MIL-STD-1472 shall also be fully explored and considered in the equipment design to minimize the possibility of failure through improper operation and maintenance and to preclude personnel safety hazards. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable functionally and from considerations of strength.

3.2.2.1 Navy design considerations. Pumps designed in accordance with this specification shall have maximum repair accessibility for ease of examination of wearing parts and for simplicity of disassembly and proper reassembly. Positioning and alignment of parts in assembly shall employ positive means such as shoulder, tongue and groove, or other locating techniques whereby correct reassembly is repeatedly assured. In consonance with the concept of ease of maintenance and where not already specified, the following features shall be considered for incorporation in the design wherever such incorporation will significantly contribute to ease of maintenance without compromising performance reliability:

a. Lifting lugs.
b. Casing assembly guide pins.
c. Alignment and positioning dowels.
d. Jacking screws.
e. Ease of access to coupling.
f. Re-makeable piping connections.

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g. Piping out of way of maintenance access.

h. Ease of packing, mechanical seal, bearing, and wearing ring replacement.

i. Split casing, bearing housings, and so forth.

j. Guards for rotating and moving components.

k. Warning plates to prevent casualties to equipment and personnel.

l. Provisions for connecting instruments for performance evaluation.

m. Attached instruments for monitoring performance.

n. Visibility and access to attached instruments.

3.2.2.2 Service life. Each pump shall have a service life of 30 years, of which 40 percent shall be actual operation. There shall be no limit on the number of starts during the life of the pump. It shall be assumed that during the life of the pump, parts subject to unavoidable wear and deterioration (with the exception of packing and seals) will be replaced at intervals no shorter than 3 years. The parts subject to wear, deterioration, and normally requiring replacement at 3-year intervals during the service life of the pump (with the exception of packing and seals) shall have a life of 10,000 actual operating hours.

3.2.2.3 Maintainability. The time to perform pump maintenance action requiring part removal shall not exceed 3 hours.

3.2.2.4 Parts interference. In order to ensure comparability of parts, calculations of the worst-case cumulative effect of dimensions and tolerances for parts and location shall be prepared. These calculations shall demonstrate:

a. There will be no interferences between rotating and stationary parts as assembled.

b. There will be no interference between rotating and stationary parts, and hence no reduction in life of wearing parts, when these worst-case cumulative effects are combined with operational, environmental, and thermal loading.

3.2.3 Shock and vibration.

3.2.3.1 Shock. Pumps and motors shall withstand the grade A shock requirements in accordance with MIL-DTL-901, as specified in 4.5.6.

3.2.3.2. Bolting. Bolts designed to be stressed in shear shall be installed in holes with a minimum of clearance. Hole diameters shall be not more than 1/32 inch larger than the bolts for sizes up to and including 3/4 inch, and no more than 1/16 inch larger than bolts of greater than 3/4 inch in size. Mounting bolts for fastening the equipment shall be in accordance with MIL-DTL-1222, grade 2 or better.

3.2.3.3 Vibration. Pumps shall not be damaged or caused to malfunction by either the environmental vibrations or by internally excited vibrations. Unless otherwise specified (see 6.2), units furnished under this specification shall withstand the type I and II vibration requirements in accordance with MIL-STD-167-1, type I and II (see 4.5.7).

3.2.4 Alignment.

3.2.4.1 Design considerations. The design of the pump units shall be such as to permit handling, shipment, and installation of the units on board ship without disturbing alignment and such that the normal distortion, warping, or vibration of the supporting structures on board ships cannot cause fracture or appreciable distortion.

3.2.4.2 Alignment design. Alignment between separate components of a unit shall be maintained by means of keyways, rabbeted or tongue and groove joints, fitted bolts, or other means. Parts manufactured to the same drawings shall be interchangeable.

3.2.4.3 Support. In no case shall a piece of equipment be rigidly supported from more than one plane.

3.2.5 Noise. The design, construction, and workmanship of the equipment shall be such that noise levels during operation at the design conditions are in accordance with the airborne noise and structureborne noise shipboard requirements of MIL-STD-1474 as specified (see 4.5.9 and 6.2).

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3.2.6 Mounting.

3.2.6.1 Installation. Pumps are normally installed to operate in a horizontal position; however, pumps shall be capable of operation and mounting in a vertical position with the motor above the pump. Pumps shall be suitable for installation either fore and aft or athwartship (see 6.2).

3.2.6.2 Dimension. Pumps shall have mounting dimensions and location of piping connections as shown on figure 1.

3.2.6.3 Mounting brackets. Mounting brackets shall contain a drip capture area for the collection of leakage from the stuffing box. The size of the tapped holes shall be as shown on figure 1. Tapped holes shall be tapered thread. Mounting brackets shall be separate from the casings and shall not include the pump stuffing box or any portion of the pressure boundary.

3.2.6.4 Close coupled assembly. Motors shall be assembled to pumps during manufacturing.

3.2.7 Bearings.

3.2.7.1 Size of radial and thrust ball bearings. The size of the radial and thrust ball bearings for each size motor frame shall be as shown on figures 2 and 3. The pump end bearing shall be fixed in the housing and shall carry the pump thrust load. However, on low noise level units, the motor end bearing may carry the thrust load. Bearings shall be selected for a B-10 life of 10,000 hours calculated in accordance with ANSI/ABMA 9. Calculations shall be made for the design point and also for the highest loading conditions over the operating range and shall consider both thrust and radial hydraulic loads, gravity loads, and bearing preloads. Hydraulic thrust due to the use of single inlet impellers without balancing rings shall be computed in accordance with the following formula:

$$T = P(A_1 - A_0) + 0.05\ P(A_2 - A_1)$$

Where:

T = Maximum unbalanced thrust in pounds.

P = Pressure in pounds per square inch (lb/in^2) at shutoff.

A_2 = Area of impeller at radius R_2 in square inches (see figure 4).

A_1 = Area of impeller to outside of impeller wearing ring, radius R_1 in square inches.

A_0 = Area in square inches to outside of shaft sleeve, radius R_0.

3.2.7.2 Excessive thrust loads. When the axial thrust load calculated by the formula of 3.2.7.1 exceeds the permissible thrust loads shown in table I, thrust balancing devices shall be added.

TABLE I. Permitted thrust loads.

Bearing Size	Revolutions per minute (r/min)			
	1,150	**1,750**	**2,800**	**3,500**
305	---	215	---	160
306	---	280	---	215
307	---	235	---	255
308	---	400	---	300
309	560	470	375	350
310	600	500	400	360
312	700	580	470	425
313	800	670	530	495

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3.2.7.3 Bearings. Motor bearings shall be in accordance with MIL-DTL-17060. The sizes used for the various frames shall be as shown on figure 3. Whenever possible, sealed bearings shall be used.

3.2.7.4 Thrust bearings. For installations utilizing the pump end bearing as the thrust bearing, a bearing preload spring shall be provided and shall be installed against the outer edge of the outer ring of the front end bearing. For installations using the front end bearing as the thrust bearing, a bearing preload spring shall be installed against the inner edge of the outer ring of the pump end bearing.

3.2.8 Lubrication.

3.2.8.1 Bearing lubrication. If sealed bearings cannot be used, then bearings shall be lubricated with DOD-G-24508 in accordance with MIL-DTL-17060.

3.2.8.2 Lubrication method. The rolling contact bearing lubrication method shall prevent bearing failure resulting from over-greasing. Bearing relubrication procedures, if required, shall be accomplished with the unit secured.

3.2.8.3 Grease lubricants. Grease lubricants shall be in accordance with DOD-G-24508. The design shall be such that any qualified grease under DOD-G-24508 can be used interchangeably with any other qualified product to the same specification. Bearings shall be capable of being relubricated without removing the bearings and shall not require frequent lubrication if the pump is not operated over an extended period of time.

3.2.8.4 Thrust bearing lubrication. Thrust bearing housings shall be so designed that the thrust bearing will be adequately lubricated immediately when the shaft starts to turn. This shall be accomplished by properly locating lubricant supply and drain fittings and by adequate sealing of the bearing housings so that, regardless of length of time the units are idle, an adequate supply of lubricant will remain in the bearing housings.

3.2.9 Packing and gasket. Packing and gaskets shall be in accordance with S9086-CM-STM-020/078. Packing shall be of the impregnated type in accordance with MIL-PRF-24790 for water or MIL-P-17416 for petroleum fluids. Gasket compression and tolerances shall not affect bearing bushing or wearing ring fits to an extent that will adversely affect reliability and performance. Pump casing joints may be made up using compressed sheet gaskets or O-rings when approved by NAVSEA.

3.2.10 Threaded fasteners. Threaded parts, such as bolts, studs, and nuts, shall be in accordance with FED-STD-H28, FED-STD-H28/2, and MIL-DTL-1222. The use of tap-bolts or cap-screws is prohibited unless approved by NAVSEA and where the use of through bolts or studs is impractical. For temperatures under 250 °F, the setting end of studs shall be a class 2 fit or preferably a class 3 fit, both with locking resin in accordance with ASTM D5363, type I, grade K. The nut end shall be a class 3 fit.

3.2.10.1 Minimum stud engagement. The length of minimum stud engagement shall be computed in accordance with FED-STD-H28.

3.2.10.2 Threaded fasteners. Unless otherwise specified (see 6.2), threaded fasteners of normally stocked lengths shall be used. Male threads on threaded fasteners, after being installed and tightened, shall protrude at least one thread beyond the top of the nut or plastic locking element. Excessive protrusion shall be avoided, particularly where necessary clearances, accessibility, and safety are important. Where practicable, the number of threads protruding shall not exceed five; however, in no case shall thread protrusion exceed 10 threads. Washers shall not be used for the sole purpose of lessening thread protrusion.

3.2.11 Piping root connections. Root connections for attaching pipes shall be in accordance with Drawing B-214.

3.2.11.1 Flanges. Pumps shall be fitted with flanged connections in accordance with MIL-STD-777 for suction and discharge pressure gauges or SAE connections in accordance with SAE J1926/1, as specified (see 6.2). All pump flanges shall have a face circular lay finish as detailed in the applicable general note of MIL-STD-777.

3.2.11.2 Pressure gauges. Pressure gauges, when provided with the units, shall be mounted in accordance with 803-1385850. Gauges shall be positioned for accessibility of reading and maintenance. Gauges shall be secured on the parent equipment in such a manner as to prevent vibration, breakage, and disconnection.

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3.2.12 Welding. Welding and welding repair shall be in accordance with S9074-AR-GIB-010/278. Brazed joints shall not be used.

3.2.13 Mercury prohibition. Pumps, drivers, and auxiliary parts furnished shall be free of mercury contamination and free of functional mercury.

3.2.14 Motors. Motors shall be AC in accordance with MIL-M-17059 for fractional horsepower (hp) or MIL-DTL-17060 for integral hp, except as otherwise specified herein, and shall conform to the following classification requirements:

Service	A
Ambient temperature	122 °F (50 degrees Celsius (°C))
Duty	Continuous
Type	Squirrel-cage induction
Design (for integral hp motors)	B
Bearings	Ball (see figures 2 and 3)
Insulation	Class B, F, H, or N
Temperature rise	158 °F (70 °C) in 122 °F (50 °C) ambient
Number of phases	3
Frequency	60 Hertz (Hz)
Voltage	440 volts
Enclosure	Totally enclosed or totally enclosed fan-cooled

3.2.14.1 Frame numbers. The standardized frame numbers for the various ratings of close-coupled pump motors shall be as specified in table II. Frame flange and shaft extension dimensions shall be as shown on figures 2 and 3, as applicable. The mounting bracket between the motor and pump volute shall be furnished.

TABLE II. AC motors 122 °F (50 °C) ambient, class B, F, H, or N insulation with 158 °F (70 °C) temperature rise (see 3.2.14).

Pump [1/]	Frame number					
	Hp	3,500 r/min	1,750 r/min	1,150 r/min	AJ [2/]	AK [2/]
1D-1750-0.33	⅓		56			
1D-3500-0.50	½	56	56			
1D-3500-0.75	¾	56				
1D-3500-1	1	182TCN			7¼	8½
1D-3500-1.50	1½	182TCN			7¼	8½
1D-3500-2	2	213TCN			7¼	8½
2D-3500-2	2	182TCN			7¼	8½
2D-3500-3	3	184TCN			7¼	8½
2D-3500-5	5	213TCN			7¼	8½

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TABLE II. AC motors 122 °F (50 °C) ambient, class B, F, H, or N insulation with 158 °F (70 °C) temperature rise (see 3.2.14) – Continued.

Pump [1/]	Frame number					
	Hp	3,500 r/min	1,750 r/min	1,150 r/min	AJ [2/]	AK [2/]
3D-1750-0.33	⅓		56			
3D-1750-0.50	½		56			
3D-3500-1	1	182TCN			7¼	8½
3D-3500-1.50	1½	182TCN			7¼	8½
3D-3500-2	2	182TCN			7¼	8½
3D-3500-3	3	184			7¼	8½
4D-1750-0.75	¾		56			
4D-1750-1	1		182TCN			
4D-3500-5	5	213TCN			7¼	8½
4D-3500-7.50	7½	215TCN			7¼	8½
5D-1750-0.75	¾		56			
5D-1750-1	1		182TCN		7¼	8½
5D-3500-2	2	182TCN			7¼	8½
5D-3500-3	3	184TCN			7¼	8½
5D-3500-5	5	213TCN			7¼	8½
5D-3500-7.50	7½	215TCN			7¼	8½
6D-1750-2	2		182TCN		7¼	8½
6D-1750-3	3		184TCN		7¼	8½
6D-1750-5	5		213TCN		7¼	8½
7D-3500-10	10	254TCN			7¼	8½
7D-3500-15	15	256TCN			7¼	8½
7D-3500-20	20	286TCN			9	10½
8D-3500-10	10	254TCN			7¼	8½
8D-3500-15	15	256TCN			7¼	8½
8D-3500-20	20	286TCN			9	10½
9D-3500-10	10	254TCN			7¼	8½
9D-3500-15	15	256TCN			7¼	8½
9D-3500-20	20	286TCN			9	10½
10D-3500-10	10	254TCN			7¼	8½
10D-3500-15	15	256TCN			7¼	8½
10D-3500-20	20	285TCN			9	10½
11D-1750-7.50	7½		215TCN		7¼	8½
11D-1750-10	10		254TCN		7¼	8½
11D-1750-15	15		256TCN		7¼	8½
12D-1150-5	5			254TCN	7¼	8½
12D-1150-7.50	7½			256TCN	7¼	8½
12D-1150-10	10			256TCN	7¼	8½
13D-1150-15	15			326TCNI [3/]	14	16
13D-1150-20	20			326TCNI [3/]	14	16
13D-1150-25	25			326TCNI [3/]	14	16

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TABLE II. AC motors 122 °F (50 °C) ambient, class B, F, H, or N insulation with 158 °F (70 °C) temperature rise (see 3.2.14) – Continued.

Pump [1/]	Frame number					
	Hp	3,500 r/min	1,750 r/min	1,150 r/min	AJ [2/]	AK [2/]
14D-3500-7.50	7½	215TCN			7¼	8½
15D-3500-5	5	213TCN			7¼	8½
15D-3500-7.50	7½	215TCN			7¼	8½
16D-1750-15	15		286TCNI [3/]		14	16
16D-1750-20	20		286TCNI [3/]		14	16
16D-1750-25	25		286TCNI [3/]		14	16

NOTES:

[1/] The Navy standard pump numbers under this specification have the letter "D" added (for example, 9D) to distinguish the pumps covered herein from those under the previous editions of this specification.

[2/] AJ and AK dimensions are shown on figure 3.

[3/] These frame sizes are larger than those shown in MIL-DTL-17060 to accommodate the larger bearing, shaft, and mounting dimensions invoked by this specification.

3.2.15 Painting. External unmachined surfaces of ungalvanized ferrous metal parts shall be thoroughly cleaned and coated with one coat of pretreatment in accordance with DOD-P-15328 and one coat of zinc chromate primer in accordance with TT-P-645, followed by a finish coat of light gray equipment enamel in accordance with MIL-DTL-15090. Painting of external surfaces of nonferrous parts of pumps is not required.

3.2.16 Identification plates.

3.2.16.1 Types. Identification plates shall be furnished on each pump and shall be type A or B in accordance with MIL-DTL-15024 and MIL-DTL-15024/5, except that identification plates of plastic or aluminum shall not be used.

3.2.16.2 Attachment. Identification plates shall be secured to equipment with corrosion-resistant metallic screws or rivets.

3.2.16.3 Contents. Pump identification plates shall include the following:

a. Manufacturer's name.
b. Manufacturer's model or type and size.
c. Service application.
d. Manufacturer's serial number.
e. Salient design characteristics:
    (1) Capacity in gallons per minute (gal/min)
    (2) Total head in lb/in^2
    (3) Speed of shaft in r/min
    (4) Brake hp
    (5) Test pressure
    (6) Special data vital to the unit:
        (a) Suction pressure
        (b) Submergence
        (c) Impeller diameter

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f. Contract number (and item number for multiple unit orders).

g. National stock number.

h. Section for Defense Contract Management Agency (DCMA) stamp.

i. Navy standard pump number (see table II).

j. Component identification number (CID) consisting of nine digits that will be supplied to the successful bidder. No materials should be shipped until the CID is included on the identification plate.

k. Nameplates to specify actual characteristics of maximum head capacity (MHC) on top half with space provided on bottom half to specify new characteristics as a result of the trimming. The following warning instructions shall be stamped in the middle of the nameplate of the pump unit:

"WARNING: The pump included in this pump unit contains a maximum diameter impeller. Before installation, verify required operating characteristics. Refer to instruction book for impeller trimming instructions. After impeller in this unit is trimmed, complete lower section of nameplate."

l. For pumps that cannot accommodate one nameplate containing the complete information required, the use of a standard nameplate on the pumps plus a bottom section nameplate for MHC pumps, mounted directly below the standard nameplate, is acceptable. A middle nameplate (for MHC pumps only) may be a separate nameplate for attaching to the side of the mounting bracket.

3.2.16.4 Additional identification plates. Each driving unit and each accessory unit shall have an identification plate in accordance with their applicable equipment specification.

3.2.17 Flanges, valves, and threaded items.

3.2.17.1 Valve flanges, fittings, and bolting. Unless otherwise specified (see 6.2), valve flanges, fittings, and bolting shall be in accordance with MIL-STD-777. All pump flanges shall have a face circular lay finish as detailed in the applicable general note of MIL-STD-777.

3.2.17.2 Grease fittings. Tapered pipe threads are acceptable for use in grease fittings located on nonmoving parts in accordance with SAE AS35411. Fittings not covered by MIL-STD-777 for application when ship connections are not involved shall be approved by NAVSEA.

3.2.18 Locking devices. Internal threaded fasteners and threaded machine elements shall be secured by locking devices where continuous operation under the condition of shock, vibration, and temperature specified herein depend upon maintaining tight connections of parts. All setscrews internal to the pressure boundary of the pump and all setscrews on internal and external rotating portions of the pump shall be secured by locking devices.

3.2.19 Reverse rotation. Pumps shall withstand reverse rotation for a period of 1 minute at all speeds up to design speed without damage.

3.2.20 Provision for handling. Lifting eyes, lugs, holes, and other means shall be prepared to permit attachment of lifting gear for lifting the assembled pump, driver, and attached accessories as a complete unit, in accordance with MIL-STD-1472. Means shall be provided for the handling of parts and components weighing 150 pounds or more that cannot be handled manually during unit overhauls and preventive maintenance inspections.

3.3 Detail requirements.

3.3.1 Performance characteristics. The performance characteristics of the pump shall be as noted on figure 1, and as specified (see 6.2). Special features, such as abrasive separators (ASs) or nonmagnetic (NM) construction, if required, shall be as specified (see 6.2).

3.3.2 Detail design characteristics.

3.3.2.1 Pump selection. Pumps for a specific application shall be selected from the rating chart shown on figure 5. The standardized pump size with the hp and motor speed shown on the rating chart for the application shall be furnished. When specified (see 6.2), pumps shall be of MHC design. MHC pumps are those furnished with an impeller of the maximum diameter that will not overload the motor at any point along the performance curve.

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3.3.2.2 Capacity curves. Pumps for all services shall have constantly rising head-capacity curves from maximum or side-open capacity to zero or shut-off capacity.

3.3.2.3 Suction lift. Pumps shall be capable of satisfactory operation at a dynamic suction lift of 15 feet (4.6 meters) of water at 85 °F (30 °C).

3.3.2.4 Motor overload. Pump units shall not overload the motor at any condition of operation from shut-off capacity to wide-open while pumping the specified fluid.

3.3.2.5 Operation. Pumps shall operate satisfactorily at any condition of operation from shut-off capacity to wide-open while pumping the specified fluid.

3.3.2.6 Rotation. Pumps shall be right-hand rotation (clockwise when viewed from the motor end of unit) only.

3.3.2.7 Wearing rings. Pumps shall be fitted with casing wearing rings which shall be secured from rotating by set screws mounted in the joint between the casing and ring.

3.3.3 Casings.

3.3.3.1 Construction. Casings shall permit the examination and replacement of wearing parts. Suction covers shall be removable to permit removal of the impeller or casing wearing ring without dismantling the unit.

3.3.3.2 Arrangement. Pumps shall have casings arranged so that the discharge may be located in any of the three positions shown on figure 1.

3.3.3.3 Bosses. Casings shall be provided with the necessary bosses for vent, drain, priming, and gauge connection in accordance with SAE J1926/1 for the thread positions of discharge. When specified (see 6.2), bosses or gauge connections shall be of sufficient size for conversion to flanged connections to socket weld type design suitable for a nipple and flanged connections in accordance with Drawing B-214.

3.3.3.4 Vent and drain holes. Casings shall be provided with a vent cock in the proper location for the position of discharge. The remaining unused vent and drain hole bosses shall be provided with plugs in accordance with MS18229.

3.3.3.5 Pressure. Casings shall withstand the applicable hydrostatic pressures shown on figure 1 (see 4.5.3).

3.3.3.6 Spare wearing rings. Spare casing wearing rings (see 6.4) shall be machined undersize by at least 0.050 inch on the diameter in order to permit finish machining during shipboard installation to mate with used impellers.

3.3.4 Suction and discharge connections. Suction and discharge connections shall be flanged for mating with flanges in accordance with MIL-STD-777.

3.3.5 Impellers.

3.3.5.1 Type. Impellers shall be of the single inlet closed type. Outside and inside surfaces of the impellers shall be smooth. Impellers shall be dynamically balanced.

3.3.5.2 Attachment. Impellers shall be keyed on the shaft and securely held against lateral movement by locked nuts or other means acceptable to NAVSEA. Shaft threading shall be counter to the direction of rotation.

3.3.5.3 Design. Impellers shall not be furnished with wearing rings. The impeller wearing surfaces shall have material thickness to permit reducing the diameter of the impeller wearing surfaces by as much as 0.050 inch to accommodate undersize casing wearing rings to restore design running clearance.

3.3.5.4 Diameter. The impeller shall have the approximate maximum diameter for each pump as specified in table III.

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TABLE III. Approximate maximum impeller diameter.

Pump number	Approximate maximum impeller diameter (inches)	Pump number	Approximate maximum impeller diameter (inches)
1D	5	10D	7½
2D	6¾	11D	9
3D	5	12D	9¾
4D	6¼	13D	10½
5D	5⅛	14D	8
6D	7	15D	5¼
7D	10	16D	9
8D	9¼	17D	10½
9D	9¼	---	---
NOTE: The Navy standard pump numbers under this specification have the letter "D" added (for example, 9D) to distinguish the pumps covered herein from those under the previous editions of this specification.			

3.3.5.5 Rotor balance. All motor rotors shall be dynamically balanced in accordance with MIL-DTL-17060. Pump impellers shall be balanced individually so that, when assembled, the pump unit unbalance is in accordance with the limits specified in MIL-STD-167-1.

3.3.6 Shafts.

3.3.6.1 Flingers. Shaft flingers shall be provided adjacent to stuffing boxes for the pumps.

3.3.6.2 Material. Shafts shall be made of nickel-copper or nickel-copper-aluminum (Ni-Cu-Al) alloy, one-piece construction.

3.3.6.3 Sleeves. Shaft sleeves shall have the dimensions shown on figure 6 (see 3.3.7) and shall suit the mechanical seal.

3.3.6.4 O-ring seals. O-ring seals shall be used between the shaft and stuffing box shaft sleeve. Other positive methods of sealing may be used subject to the approval of NAVSEA.

3.3.7 Mechanical shaft seals and stuffing boxes.

3.3.7.1 Mechanical seals. All pumps shall be provided with mechanical shaft seals in accordance with ASTM F1511 US Navy Supplement. Magnetic type seals shall not be used.

3.3.7.2 Packing gland. Each seal shall be provided with a solid gland plate capable of accommodating two or more rings of packing in accordance with MIL-PRF-24790 for water or MIL-P-17416 for petroleum fluids for use in the event of a mechanical seal failure. The gland seal shall also be provided with a flush connection to ensure positive liquid pressure is supplied to the seal faces under all operating conditions and that there is adequate circulation of the liquid at the seal faces to minimize deposits of foreign matter in the seal parts.

3.3.7.3 Metal seal materials. Mechanical seal shall be constructed out of materials in accordance with ASTM F1511 and F1511 US Navy Supplement for seawater service pumps.

3.3.7.4 Mechanical seal positioning. Mechanical shaft seals shall be positioned on the shaft by means of stub or step sleeves. Mechanical shaft seals shall not be positioned by use of set screws.

3.3.7.5 Stuffing box gland securing. Stuffing box glands shall be set up by nuts threaded on studs secured in the casings or seal gland plate.

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3.3.7.6 Seal lubrication. The pump seal utilized shall ensure that there will be adequate circulation of liquid and adequate lubrication of the seal faces when the seal is installed as intended.

3.3.8 Material. The materials of the pump shall conform to the materials specified in table IV. However, this specification is not intended to be restrictive provided proposed alternative materials will give equal or better service than the material specified. Proposed alternative materials shall be subject to approval by NAVSEA. Components of the pump for which the specific materials are not specified shall be materials best suited for the intended service. Materials that can be sensitized and are subjected to heat treatment in the sensitization range during fabrication shall be capable of passing the intergranular corrosion tests of the base material specifications. Particular attention should be given to avoiding sensitization of materials during hard facing, stress-relieving, or repair welding.

3.3.8.1 Toxicity and prohibited materials.

3.3.8.1.1 Toxicity. When evaluated in accordance with 4.6, the pump shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.6 and 6.5).

3.3.8.1.2 Prohibited materials. The pump shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2.

3.3.8.1.3 Additional prohibited materials. The following materials shall be prohibited: magnesium and its alloys, cadmium plating, asbestos, and any radioactive materials. The use of cadmium-plated parts and fasteners, including washers, is also prohibited. Lead and its compounds shall not be used where they can come in contact with seawater feed, permeate, or freshwater. The lead content in bronze alloys shall be in accordance with levels allowed by ASTM B61. For drinking water contact, all materials that may come in contact with potable freshwater shall be in accordance with NSF/ANSI 61.

3.3.8.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

TABLE IV. Materials.

Application	Material	Specification	ASTM publication
Casings	Copper-nickel alloy	---	ASTM B369, UNS C96400
Studs, plugs, adapters, bolts, nuts, and petcocks [1/]	Nickel-copper alloy	QQ-N-281, class A or B	ASTM B164, UNS N04400
	Ni-Cu-Al alloy	QQ-N-286	---
Impellers	Nickel-copper alloy	---	ASTM A494/A494M
Casing wearing rings [2/]	Nickel-copper alloy	MIL-C-24723	ASTM A494/A494M
Shafts [3/]	Nickel-copper alloy	QQ-N-281, class A or B	ASTM B164, UNS N04400 or N04405
	Ni-Cu-Al alloy	QQ-N-286, class A or B	---
Shaft sleeves, stub, or step	Ni-Cu-Al alloy	QQ-N-286, class A or B, min. Bhn 265	---
	Nickel-copper alloy	QQ-N-281, class A or B	ASTM B164, UNS N04400 or N04405

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TABLE IV. Materials – Continued.

Application	Material	Specification	ASTM publication
Flingers and mounting brackets	Gun metal	---	ASTM B584, alloy C90300 or C90500
	Copper-nickel alloy	---	ASTM B369, alloy 964 UNS C96400
	Valve bronze	---	ASTM B584, alloy C92200 or ASTM B61
Impeller nuts, washers, and keys	Nickel-copper alloy	QQ-N-281, class A or B	ASTM B164, UNS N04400 or N04405
Casing gaskets	Gasket, sheet, non-asbestos	MIL-DTL-24696	---
Tubing and nipple	Copper-nickel alloy	MIL-T-16420 (70-30) class 700	---
Fittings, silver brazing O-ring face seal type 1/	Nickel-copper alloy	QQ-N-281, class A or B	ASTM B164, UNS N04400 or N04405
	Copper-nickel alloy (70-30)	MIL-C-15726 (70-30)	---
Flanges (gauge and vent connections)	Copper-nickel alloy (70-30)	MIL-C-15726 (70-30)	---

NOTES:

1/ Mechanical flareless (bite), compression, or flared type fittings are not permitted.

2/ Wearing rings and opposing surfaces shall be of dissimilar material or shall have a difference in hardness of at least 50 Brinell hardness number (Bhn).

3/ The Ni-Cu-Al material shall be annealed prior to age hardening.

3.3.9 Part interchangeability. All parts for pumps manufactured to the same pump numbers (see table II) shall be interchangeable.

3.3.9.1 Installation interchangeability. Pumps of a given capacity in gal/min and total dynamic head in feet and type of electrical voltage shall be interchangeable in mounting dimensions, size and location of piping connections, shaft extension, shaft sleeve, and flinger.

3.4 Special tools. Special tools shall be as specified (see 6.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on the pump and its appurtenances and controls when a first article sample is required (see 3.1). This inspection shall include the examinations of 4.5.1 and 4.5.2, the tests of 4.5.3 and 4.5.5 through 4.5.9.1.2, and the assessment of 4.6, as specified in table V. The first article tests on one pump design may be conducted concurrently if practicable. Vertical and horizontal orientations of the pump shall be tested as specified (see 6.2).

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.5.1 and the tests of 4.5.3 and 4.5.4, as specified in table V. Vertical or horizontal orientations of the pump shall be tested as specified (see 6.2).

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TABLE V. Examination and tests.

Inspection	Requirement	Verification	First article	Conformance
Visual and dimensional examination	3.2 through 3.4	4.5.1	X	X
Reliability and maintainability	3.2.2.3, 3.2.20, 3.3.9, 3.4	4.5.2	X	
Hydrostatic	3.3.3.5	4.5.3	X	X
Operational	3.3.2.5	4.5.4		X
Performance	3.3.1	4.5.5	X	
Shock	3.2.3	4.5.6	X	
Vibration	3.2.3.3	4.5.7	X	
Endurance	3.2	4.5.8	X	
Noise	3.2.5	4.5.9	X	
Toxicity and prohibited materials	3.3.8.1	4.6	X	

4.4 Motor testing. Motors shall be tested in accordance with the applicable motor specification prior to assembly with the pump.

4.5 Tests.

4.5.1 Visual and dimensional examination. Each pump assembly shall be examined for compliance with the requirements specified in 3.2 through 3.4. Any redesign or modification of the standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet the specified requirements, shall be inspected for compliance with all visual examinations and dimensional measurements. Welding and castings shall be inspected in accordance with S9074-AR-GIB-010/278. Noncompliance with any specified requirements or presence of one or more defects shall constitute cause for rejection.

4.5.2 Reliability and maintainability examination.

4.5.2.1 Reliability. The pump unit shall meet the requirements for reliability as specified in 3.2.2.3, 3.2.20, 3.3.9, and 3.4. The reliability prediction methods in MIL-HDBK-217 may be used.

4.5.2.2 Maintainability. The pump unit shall meet the requirements for maintenance and repair as specified in 3.2.2.3, 3.2.20, 3.3.9, and 3.4. The maintenance procedures, removal of parts for repair, and the time required for accomplishment shall be verified during conformance testing. The demonstration procedures in MIL-HDBK-470 may be used.

4.5.3 Hydrostatic test. Each pump assembly shall be tested hydrostatically to pressures shown on figure 1. Hydrostatic test pressures shall be maintained for at least 15 minutes, or longer, as necessary for examination of the entire casing. The casing and suction cover shall exhibit no leakage through the casing material or joints.

4.5.4 Operational test. Each pump assembled with its driving motor shall be tested at its rated operating speed and point of rating by a continuous non-stop run of at least 30 minutes to check operation and smoothness of running.

4.5.5 Performance tests.

4.5.5.1 Test procedures. Performance tests shall be in accordance with HI 40.6, to include creation of a flow map, and conducted under laboratory conditions therein.

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4.5.5.2 Pumps to be tested. One pump of each size of a given contract lot shall be tested to determine the overall power consumption of the unit under the specified test conditions (see 4.5.6.3 and 6.2).

4.5.5.3 Test criteria. Performance tests shall demonstrate the ability of the pump to handle its rated capacity of specified liquid at the maximum temperature and minimum suction head, or maximum suction lift or vacuum, as applicable. For MHC pump, these tests shall be performed with the maximum diameter impeller allowed for the size of the pump being tested.

4.5.5.4 Pump curves. The following test curves shall define the rated speed conditions of the motor. Head capacity and brake hp curves shall demonstrate capacities from over-loaded at any point from shut-off to wide open.

a. Total head versus capacity.

b. Pump efficiency versus capacity.

c. Brake hp versus capacity.

4.5.5.5 Additional motor testing. Unless previously performed, one pump of each size shall also be given additional tests using dynamometer or calibrated motor (motor manufacturer's efficiency test may be used to identify motor calibration) to define the following curves:

a. Gal/min versus total dynamic head at constant rated r/min for various impeller diameters with superimposed constant efficiency and constant brake hp curves at 1.00 specific gravity. The number of impeller diameters shall be sufficient to develop a family of curves.

b. Determination of the maximum suction lift in feet versus gal/min at the maximum impeller diameter tested.

4.5.6 Shock test.

4.5.6.1 Unit under test. Unless otherwise specified (see 6.2), one complete pumping unit of each size (pump and motor) shall be shock tested in accordance with MIL-DTL-901, grade A. Pump units shall be shock tested running and at standstill on alternate blows.

4.5.6.2 Retest of designs already tested. Equipment exactly duplicating that previously shock tested and accepted need not be retested except when evidence of low shock resistance develops in installed units or NAVSEA determines additional shock tests are necessary.

4.5.6.3 Test conditions. Shock mounts (resilient mounts for shock attenuation) shall not be used. All pumps shall pass the specified shock test solidly mounted, regardless of whether noise attenuation mounts are specified for mounting the unit in the ship.

4.5.6.4 Location of test. If a manufacturer is unable to conduct these tests at their own plant, they may be conducted at a commercial laboratory or Government laboratory. When shock tests are conducted at a Government laboratory, applicable drawings shall be as specified (see 6.2).

4.5.6.5 Rejection criteria. Evidence of any of the following shall be cause for rejection:

a. Breakage of parts, including mounting bolts.

b. Appreciable distortion or dislocation of a part, such as shaft, mounting feet, and bearings.

c. A mechanical unbalance of more than two times the amplitude of unbalance measured prior to test at rated speed, or more than that permitted by the specifications for the driver. The amplitude of unbalance both before and after the shock tests and the speed at which measured shall be recorded (see 6.2).

4.5.6.6 Recondition of shock tested units. Pump units which have passed the shock test, and have had their bearing replaced, may be used subject to NAVSEA approval (see 6.2).

4.5.7 Vibration test. Unless otherwise specified (see 6.2), pumps shall be tested in accordance with MIL-STD-167-1, type I and II.

4.5.8 Endurance test. The pump furnished for initial qualification shall be operated for a period of not less than 500 hours of actual running time with a minimum of 100 starts to ascertain reliability of performance and operation.

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4.5.8.1 Pre-test examination. Prior to commencement of the endurance test and immediately after completion of the endurance test, the pump shall be disassembled to the extent necessary and the critical dimensions and running clearance of parts subject to wear, erosion, and derangement shall be measured and calculated. Components such as pump impellers and casings subject to erosion, corrosion, cavitation, and wear, the effects of which are not subject to routine measurement, shall be reexamined for their condition after completion of the test.

4.5.8.2 Endurance capacity verification. During the initial and final hours of the endurance test run, the capacity test in accordance with 4.5.5 shall be performed to determine the changes in the pump's performance characteristics.

4.5.8.3 Conduct of endurance test. The endurance test shall not be continuous but shall be interrupted by at least three rest periods of approximately 8 hours each. The number of starts specified in 4.6.8 at full line voltage shall be performed during the course of the test. During an early part of the endurance test, the pump shall be operated continuously for 24 hours at a capacity as near free delivery as possible at maximum rated speed and normal specified temperature, submergence, and suction conditions. During the latter part of the endurance test, the pump shall be operated as near shut-off as possible for 12 hours continuously. The remainder of the endurance test shall be run at maximum rated speed and within +20 °F, -0 °F of maximum specified liquid temperature. The pump shall be operated at one third, two thirds, and rated capacity in approximately equal time intervals. Operations shall be at minimum specified net positive suction head available, or maximum specified suction lift or vacuum, as applicable.

4.5.8.4 Test observation. The pump shall be monitored during the endurance test to accurately determine the conditions of operation, speed at which operated, and the general performance observed to include the following:

a. The pump seal leakage rate.

b. The smoothness of operation (normal-abnormal).

c. Other abnormal findings.

d. Adjustments made.

e. Changes made in the conditions or method of operation.

4.5.8.5 Acceptance criteria. Endurance test acceptance criteria is as follows:

a. The head-capacity curve at maximum rated speed after 500 hours of pump operation shall conform to the specification requirements and shall show no abnormal deviations from the curve before the 500-hour test.

b. The unit performance and operation after 500 hours of operation shall be unchanged and normal and meet all specification requirements.

c. The unit operation at the end of the endurance test shall be smooth and in conformance with the specification.

d. The leakage rate from a mechanical seal shall not exceed 5 drops per minute.

e. Running clearances shall be normal.

f. Components subject to attack from corrosion, erosion, and cavitation shall be in a condition commensurate with 500 hours of service.

g. Wear rates for wearing parts critical for proper operation shall demonstrate a rate of wear for the test period that shall be consistent with the specified design life requirements.

h. No failures shall have occurred throughout the 500-hour test.

4.5.8.6 Post endurance test procedure. The unit subjected to the 500-hour endurance test shall be restored to the as-new condition by replacement of all parts worn beyond the as-new design tolerances. The restored unit shall pass the tests specified in 4.5.3, 4.5.4, and, if applicable, 4.5.9.

4.5.9 Airborne and structureborne noise tests. When specified (see 6.2), pumps shall undergo airborne and structureborne noise tests conducted in accordance with MIL-STD-1474. Noise tests shall be performed with the driver furnished with the pump.

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4.5.9.1 Acceptance criteria.

4.5.9.1.1 Airborne noise. The unit shall be in accordance with the airborne noise level limits specified in MIL-STD-1474. If the first three production units are 10 decibels (dB) lower in all octave bands than those specified in MIL-STD-1474, and providing all production units are in accordance with the balance requirements of MIL-STD-167-1, type II or the structureborne noise levels specified (see 6.2), then airborne noise testing need not be conducted on the remaining units.

4.5.9.1.2 Structureborne noise. When required, the unit shall be in accordance with the structureborne noise level limits specified (see 6.2). Unless otherwise specified (see 6.2), specific structureborne noise levels shall be in accordance with MIL-STD-1474, type II.

4.6 Toxicity and prohibited materials. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.3.8.1.1 and 3.3.8.1.2, as specified (see 6.2). The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the pumps using data provided by the manufacturer/distributor to the NMCPHC (see 3.3.8.1.1 and 6.5).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Pumps covered by this specification are intended for miscellaneous services on board naval ships. Current stock of Navy standard close couple centrifugal pumps may be used until depleted; future requisitions for these pumps should refer to the current version of this specification. Pumps procured to this specification are form, fit, and function replacements for prior revisions.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Designation of manufacturer (pump or driver) responsible to the purchaser for compatibility of pump and driver and operation of the assembled unit.

c. Pump number (see figure 5).

d. When first article is required (see 3.1).

e. Space and weight requirements (see 3.2).

f. Performance characteristics (see 3.2 and 3.3.1).

(1) Rated capacity – gal/min or MHC.

(2) Rated total dynamic head – lb/in^2 or MHC.

(3) Liquid to be pumped.

(4) Specific gravity of liquid to be pumped.

(5) Temperature of liquid to be pumped.

(6) Suction head.

g. If vibration test requirements are not necessary (see 3.2.3.3 and 4.5.7).

h. Noise level limits and whether noise test is required (see 3.2.5 and 4.5.9).

i. Mounting of units (see 3.2.6.1).

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j. If threaded fasteners are other than as specified (see 3.2.10.2).

k. Flanged connections requirements (see 3.2.11.1).

l. Requirements for valve flanges, fittings, and bolting if not as specified in MIL-STD-777 (see 3.2.17.1).

m. Special features, if required (see 3.3.1).

n. Whether MHC design is applicable (see 3.3.2.1).

o. Whether bosses or gauge connections are required (see 3.3.3.3).

p. Special tools required (see 3.4).

q. Requirements for an HHA (see 3.3.8.1 and 4.6).

r. Orientation of testing (see 4.2 and 4.3).

s. Additional test conditions (see 4.5.5.2).

t. Shock test requirements (see 4.5.6.1, 4.5.6.5, and 4.5.6.6).

u. Requirement for copies of applicable drawings if shock tests are conducted at a Government laboratory (see 4.5.6.4).

v. Packaging requirements (see 5.1).

6.3 Recommended documentation. The following list of documentation is recommended to be requested as part of the acquisition of any pump unit(s) under this specification:

a. Engineering drawings.

b. Materials listing.

c. Pump curves.

d. Installation drawings.

e. Technical Manual.

f. Test reports.

g. Certifications.

h. On board repair parts list.

i. Alignment procedures.

j. Technical repair standard.

6.4 On board repair parts. The repair parts listed in table VI are shown for guidance only in preparing provisioning technical documentation.

TABLE VI. On board repair parts.

Name of part	Quantity per set
Ball bearings Felt wiper rings or oil seals Packing, if special Mechanical seal Shaft sleeves	100% complete replacement for all installed units per ship.
Undersize casing wearing rings Stuffing box throat bushings Impeller nuts	100% replacement for one pump per ship.

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6.4.1 Repair parts list. The list should be so treated as to be resistant to oil, water, and fading. The format of the list should include a heading and columns of data for the items listed. The heading should include titles and applicable entries as follows:

a. Reproduced from drawing number
b. Number of ships
c. Application
d. Contract number
e. Manufacturer
f. Quantities are for units per ship
g. The columns should include:
    (1) Piece number
    (2) Name of part
    (3) Quantity
    (4) Drawing number
    (5) Federal stock number

    Additional columns may be used as applicable.

6.5 Toxicity evaluation. The NMCPHC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCPHC to the Government for evaluation. The HHA process is described on the NMCPHC's website, http://www.med.navy.mil/sites/nmcphc/industrial-hygiene/Pages/health-hazard-assessment.aspx.

6.6 Subject term (key word) listing.

Head

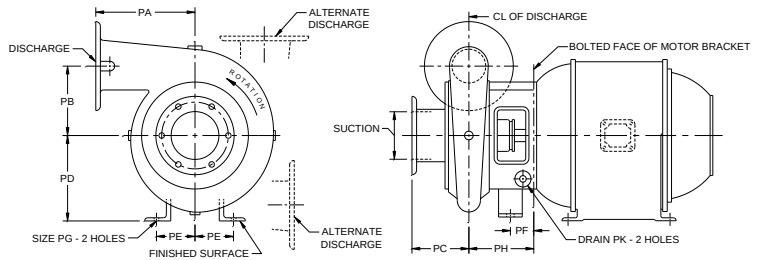
Mechanical seal

Pressure

Sealed bearing

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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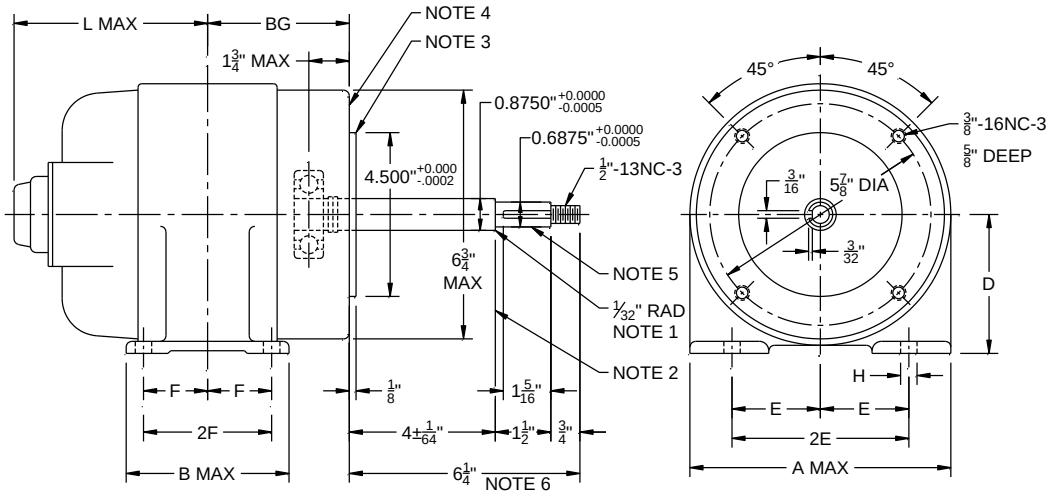


Pump No.	Navy size flange		PA	PB	PC	Hydrostatic test Press	PD	PE	PF	PG	PK	Motor frame sizes (AC)	PH	Group size shaft sleeve and water flinger
	Disch.	Suction												
1D	1	1¼	4¾	3	3¼	75	3½	2	2	17/32	⅜	56	5⅞	S
							4¾	2½	2⅜	17/32	½	182TNC	6¼	S
2D	1	1¼	5¼	3¾	3¼	125	4¾	2½	2⅜	17/32	½	182TNC-184TNC-213TNC	6¼	S
3D	1¼	1½	4¾	3⅛	3½	75	3½	2	2	17/32	⅜	56	5⅞	S
							4¾	2½	2⅜	17/32	½	182TNC-184TNC	6¼	
4D	1½	2	5½	3¾	3¾	100	4¾	2½	2⅜	17/32	½	56-182TNC-213TNC-215-TNC	6¼	S
5D	2	2	5¼	3¼	4	75	4¾	2½	2⅜	17/32	½	56-182TNC-184TNC	6¼	S
												213TNC-215TNC		
6D	3½	3½	7	5	4½	50	4¾	2½	2⅜	17/32	½	182TNC-184TNC-213TNC	6¼	S
7D	1¼	2	6¾	5¼	4	250	7	4	4	21/32	¾	254TNC-256TNC-286TNC	8 3/16	M
8D	1½	2½	6½	5	4	250						254TNC-256TNC-286TNC		
9D	2½	3½	8	5¼	4¼	250						254TNC-256TNC-286TNC		
10D	3½	5	7½	4½	6	125						254TNC-256TNC-286TNC		
11D	5	5	8½	5½	6	100						215TNC-254TNC-256TNC		
12D	6	6	9½	8	6	50						254TNC-256TNC		
*13D	8	8	10½	10	8½	50	14	4	15¾	25/32	¾	326TNCI	9½	L
14D	1	1¼	6	4¼	3½	200	4¾	2½	2⅜	17/32	½	215TNC	6¼	S
15D	2½	3	6	3¾	4½	75	4¾	2½	2⅜	17/32	½	213TNC-215TNC	6¼	S
*16D	6	8	9½	8	8½	100	14	4	14½	25/32	¾	286TNCI	9½	L
17D	2½	3	8	5½	4¾	250	7	4	4	21/32	¾		8 3/16	M

*NOTE: Supporting feet part of volute casing or cover.

FIGURE 1. Outline dimensions for Navy standard close-coupled pumps.

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SH 2291A

Standard Frame Dimensions and Bearing Sizes for Fractional Horsepower Close-Coupled Motors (Dimensions in Inches)

Frame No.	A Max.	B Max.	D	E	F	H	BG	Front End Brg. (Min.)	Pump End Brg. (Min.)	L Max.	Max WT LB
56	6¾	11½	3½	2 7/16	1½	17/32	4 1/16	303	305	9¼	55

NOTES:

1. Radius shall be smooth and center locus concentric with shaft axis.
2. Sealing surface shall be smooth and perpendicular to shaft axis.
3. Diameter to be concentric with shaft at impeller location within 0.004 inch total indicator reading.
4. Face of flange to be square with shaft within 0.004 inch total indicator reading.
5. Runout of shaft 0.002 inch maximum total indicator reading.
6. Total axial end-play of shaft in assembled motor at no-load shall not exceed 0.010 inch. Bearing pre-load springs may be used to decrease end-play.

Tolerances:

D Dimension Frame 56 ---------- +0", -1/32"
2E and 2F ---------- +1/64", -1/64"
BG Minus F ---------- +1/32", -1/32"
Keys
Width 3/16" (0.188) ---------- +0.000", -0.002"
Depth

Shaft Diameter	Key Dimensions Width	Depth	Bottom of Keyway to Opposite Side of Shaft
0.6875	3/16	× 3/16	0.580 - 0.565

Keyways 0.188 ---------- +0.001", -0.001"

FIGURE 2. Standard dimensions of fractional hp close-coupled motors.

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Tolerances:

D Dimension

Frames 182 TCNI - 326 TCNI ---------- +0", -1/32"

2E and 2F ---------- +1/64", -1/64"

BG minus F ---------- +1/32", -1/32"

Keys

Width ---------- +0.000", -0.002"

Depth

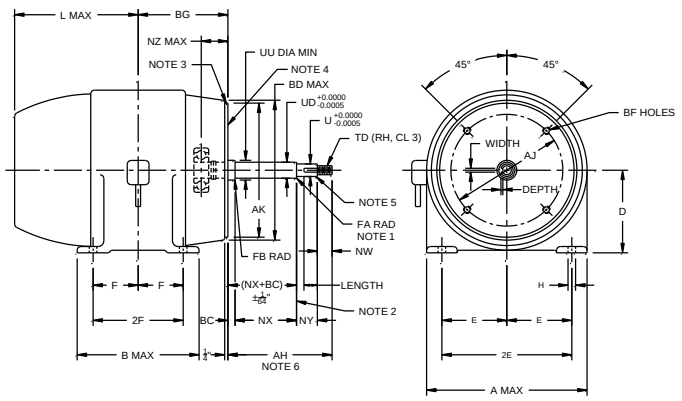
Shaft Diameter	Key Dimensions Width		Depth	Bottom of Keyway to Opposite Side of Shaft
0.6875	3/16	×	3/16	0.580 - 0.565
1.0000	1/4	×	1/4	0.859 - 0.844
1.3750	3/8	×	3/8	1.161 – 1.146

AK Dimension

10.5" and Smaller ---------- +0.000", -0.002"

12.5" and Larger ---------- +0.000", -0.003"

Keyways ---------- +0.001", -0.001"



8h 2991A

NOTES:

1. Radius shall be smooth and center locus concentric with shaft axis.
2. Sealing surface shall be smooth and perpendicular to shaft axis.
3. Diameter to be concentric with shaft at "U" dimension within 0.004 inch total indicator reading.
4. Face of flange to be square with shaft within 0.004 inch total indicator reading.
5. The shaft runout shall be as follows:

Frame Size	Shaft Runout (TIR)
182 TCN thru 286 TCNI	0.002 inch max.
326 TNCI	0.003 inch max.

6. Total axial end-play of shaft in assembled motor at no-load shall not exceed 0.010 inch for 0.310 or smaller bearings and 0.015 inch for bearings larger than 310. Bearing pre-load springs may be used to decrease end-play.
7. All dimensions shown on figure 3 comply with MIL-DTL-17060, except for frames 286 TCNI and 326 TCNI which require larger bearings, shafts, and mounting bracket.
8. The "O" ring mating surface on the shaft shall have a 32 finish (see figure 6, detail A).

Standard Frame Flange and Shaft Extension Dimensions and Bearing Sizes for Integral Horsepower Close-Coupled AC Motors (Dimensions in Inches)

	Frame No.								
	182 TCN	**184 TCN**	**213 TCN**	**215 TCN**	**254 TCN**	**256 TCN**	**286 TCN**	**286 TCNI**	**326 TCNI**
A MAX	9	9	10½	10½	12½	12½	14	18	18
B MAX	6½	7½	7½	9	10¾	12½	14	15½	15½
D	4½	4½	5¼	5¼	6¼	6¼	7	9	9
E	3¾	3¾	4¼	4¼	5	5	5½	7	7
F	2¼	2¾	2¾	3½	4⅛	5	5½	5⅝	5⅝
H	17/32	17/32	21/32	21/32	25/32	25/32	25/32	1 1/16	1 1/16
BG	5⅞	6⅜	7¼	8	9⅛	10	10½	11¾	11¾

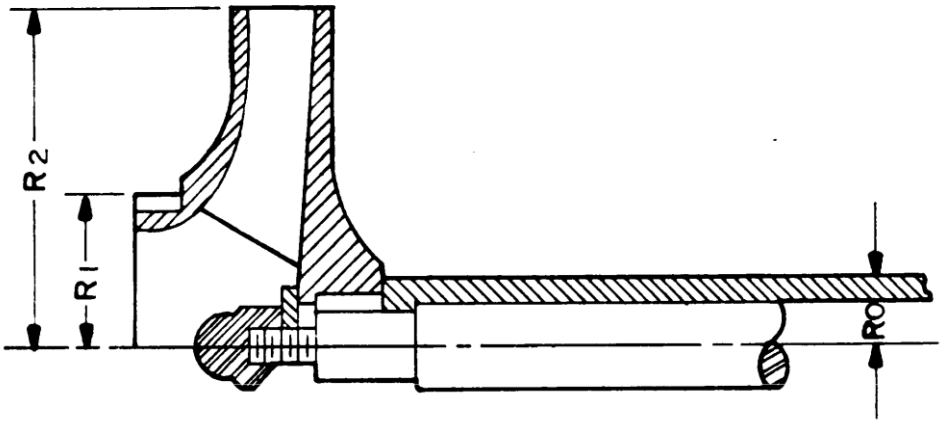
FIGURE 3. Standard dimensions and bearing sizes for integral hp close-coupled AC motors.

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		Frame No.								
		182 TCN	184 TCN	213 TCN	215 TCN	254 TCN	256 TCN	286 TCN	286 TCNI	326 TCNI
FRONT END BRG (MIN)		305	305	305	305	306	306	308	309	309
PUMP END BRG (MIN)		307	307	308	308	310	310	312	313	313
BD MAX		9	9	9	9	10	10	13	16½	16½
AJ		7¼	7¼	7¼	7¼	7¼	7¼	9	14	14
AK		8½	8½	8½	8½	8½	8½	10½	16	16
BC		3/8	3/8	3/8	3/8	11 1/16	11 1/16	11 1/16	0	0
AF	TAP	½ - 13	½ - 13	½ - 13	½ - 13	½ - 13	½ - 13	5/8 - 11	5/8 - 11	5/8 - 11
	DEPTH	¾	¾	¾	¾	¾	¾	1	1	1
AH		6 5/8	6 5/8	6 5/8	6 5/8	9 9/16	9 9/16	9 9/16	10¾	10¾
NW		¾	¾	¾	¾	1	1	1	1¼	1¼
NY		1½	1½	1½	1½	2¼	2¼	2¼	3	3
NX		4	4	4	4	5 5/8	5 5/8	5 5/8	6½	6½
TD		½ - 13	½ - 13	½ - 13	½ - 13	5/8 - 11	5/8 - 11	5/8 - 11	7/8 - 9	7/8 - 9
U		11/16	11/16	11/16	11/16	1	1	1	1 3/8	1 3/8
UD		7/8	7/8	7/8	7/8	1¼	1¼	1¼	1¾	1¾
FA		1/32	1/32	1/32	1/32	1/32	1/32	1/32	1/16	1/16
FB		¼	¼	¼	¼	¼	¼	3/8	¼	¼
KEYWAY	WIDTH	3/16	3/16	3/16	3/16	¼	¼	¼	3/8	3/8
	DEPTH	3/32	3/32	3/32	3/32	1/8	1/8	1/8	3/16	3/16
	LGTH	1 5/16	1 5/16	1 5/16	1 5/16	2	2	2	2¾	2¾
KEY SQ. END		1¼	1¼	1¼	1¼	1 15/16	1 15/16	1 15/16	2 5/8	2 5/8
NZ MAX		1 5/8	1 5/8	1 7/8	1 7/8	2¼	2¼	2 3/8	3 11/16	3 11/16
UU MIN		1.1875	1.1875	1.375	1.375	1.750	1.750	2.000	2.000	2.000
L MAX		9	9½	10¼	11	13	14	15½	17	17
MAX WGT LBS		105	130	170	205	305	370	525	970	970

FIGURE 3. Standard dimensions and bearing sizes for integral hp close-coupled AC motors - Continued.

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SH 1319914

FIGURE 4. Outside of shaft sleeve.

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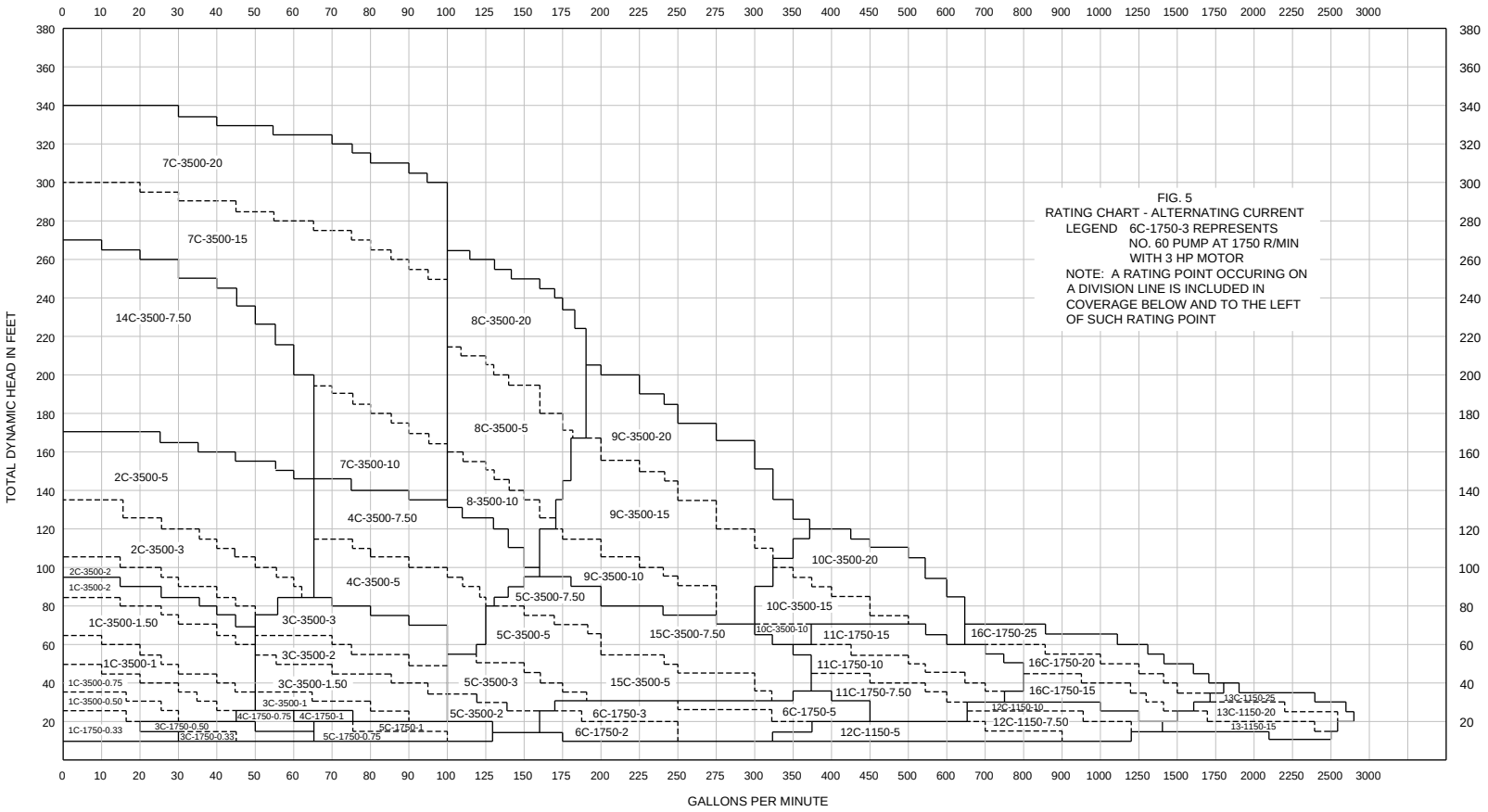
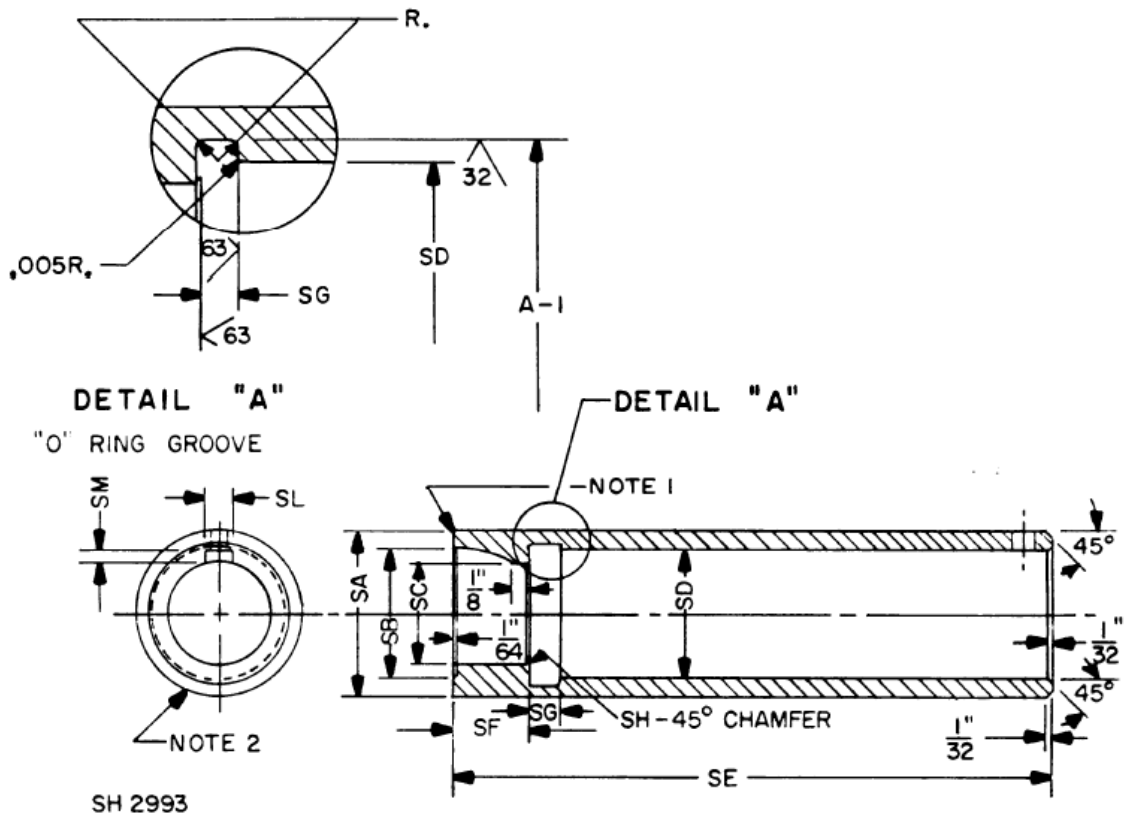


FIGURE 5. Rating charts for AC motors.

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GROUP	SA +.000 -.001	SB	SC +.002 -.000	SD +.001 -.000	SE	SF +.005 -.005	SG	SH	SL	SM	SR	ST	R	+.002 -.000 A-1	"O"-RING SIZE
S	1.125	15/16	.690	.8760	AS REQUIRED	.500	.093 .098	1/32	3/16	3/32	3/16	3/32	.005 .015	.975	2-020
M	1.625	1 5/16	1.003	1.2510		.625	.140 .145	1/32	1/4	1/8	1/4	13/64	.005 .015	1.412	2-124
L	2.250	1 7/8	1.378	1.7510		.750	.187 .192	1/16	3/8	3/16	5/16	1/4	.010 .025	1.972	2-224

NOTE 1: SEALING SURFACES SHALL BE SMOOTH TRUE AND PERPENDICULAR WITH BORE
NOTE 2: OUTSIDE DIAMETER SHALL BE GROUND SMOOTH TRUE AND CONCENTRIC WITH BORE

FIGURE 6. Shaft sleeve detail.

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CONCLUDING MATERIAL

Preparing activity:
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(Project 4320-2018-001)

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